Akanksha Sachan

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Research Interests

Epigenetic reprogramming and aging; generative and mechanistic modeling of single-cell dynamics

Education

University of Pittsburgh, Aging Institute + Dept. of Immunology, School of Medicine *'24 - Present*
- PhD in Computational Biology | Advisors: Dr. Aditi Gurkar & Dr. Jishnu Das

Carnegie Mellon University, Comp-Bio Department, School of Computer Science *'22 - '24*
- PhD in Computational Biology | Switched-Advisor | GPA: **3.58/4**

IIT Bombay, Department of Chemical Engineering, Mumbai, India *'18 - '22*
- Bachelor of Technology in Chemical Engineering | GPA: **8.06/10**

Awards and Scholarships

- **1 of 10** recipients of American Federation for Aging Research (AFAR) Graduate Scholarship *'25*
- **Best Poster** Award at the Aging Institute Research Day, School of Medicine, UPitt *'24*
- **1st Place, Student A100 track** NeurIPS LLM Efficiency Challenge | team of 5 *'23*
- **$500 Track Prize** (Healthcare Accessibility for Underrepresented Communities), Pitt Challenge Hackathon (6th annual) | team of 3 | Pitch Deck *'22*
- Recipient of Undergraduate Research Award (URA-01) by IIT Bombay *'22*
- Recipient of Merit-cum-Means (MCM) Scholarship by IIT Bombay for two semesters *'21*
- Awarded a full scholarship **($5,000)** for iSURE program by the University of Notre Dame *'20*
- **Top 0.2%** (Rank 3188 out of **1.1M+** candidates) in JEE (Main) *'18*
- **Top 1.5%** (Rank 2517 out of **160k+** candidates) in JEE (Advanced) *'18*

Invited Talks and Posters

- **"FIREFate: Functional and Interpretable Regulatory Encodings of cellular Fate"** | *paper-in-prep*
  - American Association of Immunologists (AAI) Annual Meeting 2026 | Oral *Boston, MA, Apr '26*
  - Center of Lung Aging and Regeneration, School of Medicine, UPitt | Talk *Nov '25*
  - 30th Annual School of Medicine Graduate Student Research Symposium, UPitt | Talk *Oct '25*
  - **Keystone Symposia**: AI in Molecular Biology | Oral *Santa Fe, NM, Sep '25*
  - Joint CMU-UPitt Ph.D. Program in Computational Biology, Orientation Retreat | Talk *Aug '25*
  - **ISMB/ECCB** 2025 (Regulatory and Systems Genomics track) | Oral *Liverpool, UK, Jul '25*
- **"Molecular underpinnings of accelerated skeletal muscle aging"** | *paper-in-prep*
  - Center for Systems Immunology Annual Retreat, UPitt | Talk *Nov '25*
  - 23rd Annual Department of Immunology Scientific Retreat, UPitt | Poster *Wheeling, WV, Sep '25*
  - Aging Institute Research Day, UPitt | Poster *'24*
- **"Micro-MPA digital twin, with sclerostin-dependent RANKL production reproduces pivotal dual-action effect of romosozumab found in clinical trials"** | *Publication*
  - 50th European Calcified Tissue Society (ECTS) Congress | Poster *Liverpool, UK, Apr '23*

Athletic Accolades

- **Bronze Medal** | 35th Intercollegiate (inter-IIT) Aquatics Meet | 3rd position in Medley-Relay *'19*
- **Swam 12-hours** | Swimathon organized by the IIT Bombay aquatics committee *'19*
- **Ran 26.2 miles** | Dick's Sporting Goods Pittsburgh Marathon | 5 hours 15 mins *'24*
- **Ran 13.1 miles** | Dick's Sporting Goods Pittsburgh Half Marathon | 2 hours 30 mins *'23*
- **3rd Position** | Computational Biology Department, Pretty Good Race (5km) for SCS, CMU *'23*

Leadership & Societal Contributions

- **Vice President** | *Allegheny Science Policy and Governance (ASPG)* *Aug '25 - Present*
  - Leading a graduate-student organization (recognized chapter of the **National Science Policy Network**) bridging the advocacy gap between scientists and policy-makers
  - Organized panel discussions with Representatives & Day on the State Capital Hill, Harrisburg, PA for interactions with Senators and discussions on policies for **AI/Data Centers and Aging**
- **Teaching Assistant** | *02-760: Lab Methods for Computational Biologists, SCS, CMU* *Fall '23 - Spring '24*
  - Taught a lecture on **genomics technologies** and gene regulation to a class of around 20 students
  - Revamped a module of the course based on **RNA-sequencing** analysis of mouse cell lines
- **Coordinator** | *Technights, Carnegie Mellon University, Robotics Institute* *Aug '23 - Aug '25*
  - Led a **computer-science outreach** program teaching middle-school girls technical concepts
  - Developed and taught sessions on data analysis, foundation models, and deep learning
- **Team Lead** | *4D Nucleome Consortium Hackathon, Seattle, Washington* *Mar '24*
  - Led a team of 5 during the hackathon to create integrative pipelines for **single-cell HiC** data analysis using deep-learning tools and publicly available cell-atlas scale datasets
  - Delivered a 30-minute presentation to showcase results to consortium researchers
- **Treasurer • Senator** | *Graduate Student Association, (CPCB GSA)* *Aug '23 - Aug '25*
  - Managed a budget of **$5,121** and tracked spending spanning both CMU and UPitt GSA events
  - Organized the 2024 open-house event for visiting prospective students admitted to CPCB
- **Editorial Board Member • Design Convenor** | *Insight, the student newsletter of IIT Bombay* *'19 - '21*
  - Sketched & Illustrated **1 Flagship & 1 Special Edition Magazine Cover**, and a merch t-shirt
  - Editorial Board member ('20-'21): contributed to editorial oversight and article curation
- **Convenor, Girls Swim Team** | *Aquatics, Indian Institute of Technology Bombay* *'19 - '20*
  - Led logistics, training coordination, and community building events for the institute girls' team

Consortiums & Workshops

- **SenNet Consortium Annual Meeting** | *Cellular Senescence Network, NIH, Washington, DC* *Sep '25*
  - Attended talks on cellular senescence atlasing and organ-specific senescence signatures
- **Grace Hopper Celebration** | *AnitaB.org, Philadelphia* *Oct '24*
  - Experienced talks and brainstorming sessions with tech industry leaders
- **Computational Genomics Summer Institute** | *University of California, Los Angeles* *Jul '23*
  - **Delivered Lightning Talk**: "Connecting 3D genomic organization to gene expression"
- **Grad Cohort for Women** | *Computing Research Association (CRA-WP), San Francisco* *April '23*
  - Participated in mentoring sessions from women in CS across industry and academia

Research Experience Prior to Thesis

- **Mapping the epigenetic landscape of single-cells using 3-D chromatin architecture** *'22 -'24*
  Supervisor - Prof. Jian Ma, Computational Biology Department, School of Computer Science, CMU
  - Developed a framework to identify cis-regulatory, 3-D genomic hubs, controlling target-gene expression and key transcription factors in healthy and cancerous cell lines
  - Implemented **hypergraph-based representation learning** models for integrative analysis of 3D genome architecture in single-cell, multi-omic, mouse brain datasets
- **Translational Mechanobiology Lab**, *MBI, National University of Singapore* *'22*
  Summer Internship | Supervisor - Prof. Hanry Yu, Dept. of Physiology, YLL School of Medicine
  - Implemented a **Generative-Adversarial-Network** (GAN) pipeline for colorectal cancer imaging data to support robust deep-learning deployment in clinical environments
  - Reviewed domain adaptation and distribution-shift methods for liver fibrosis staging from annotated medical images
- **Lab for Bone Biomechanics**, *Institute of Biomechanics, ETH Zurich* *'21*
  Summer Internship | Supervisor - Prof. Ralph Muller, Dept. of Health Sci&Tech | Presentation
  - Added a **Type-2 Diabetes** pathway to an agent-based **osteoporosis** model and ran large-scale simulations on the **CSCS Swiss National Supercomputing Center Cluster**
  - Validated the model against in-vivo data from 235 diabetic patients and identified osteoid production as a key parameter via **sensitivity analysis**
- **Multicellular Systems Engineering Lab**, *University of Notre Dame* *'20*
  Summer Internship | Supervisor - Prof. Jeremiah Zartman, Chemical & Biomolecular Engineering | Poster
  - Built a morphometric feature-extraction pipeline for *Drosophila* wing imaginal discs using OpenCV and Fiji to understand organism-wide signaling patterns
  - Evaluated variance under gene-knockout in the Ca^{2+} signaling pathway using PCA-features

Key Coursework

- **Machine Learning & Statistics:** Intermediary Statistics (CMU: 36-705), Probabilistic Graphical Models (CMU: 10-708), Introduction to Machine Learning (CMU: 10-701)
- **Predicting genomic-hub mediated TF-perturbation response via geometric deep learning** *'23*
  10-708: Probabilistic Graphical Models | Code Repo
  - Augmented the GEARS **GNN** for Perturb-seq response prediction with a Hi-C contact graph (K562 **cancer cell-line** Micro-C bulk-seq dataset, 20kb (base-pair) bins mapped to gene TSSs), adding 3D-genome topology to the existing co-expression and Gene Ontology graph features
- **Single-cell DNA methylation prediction using single-cell HiC data and Graph NNs** *'23*
  02-710: Computational Genomics | Code Repo
  - Developed a **GNN** to predict methylation from HiC data, connecting structure-to-phenotype
  - Performed clustering with K-means and t-SNE on **single-nucleus DNA methylation** data
- **Efficient Online Continual Semi-Supervised Learning** *'22*
  10-701: Intro to ML
  - Improved semi-supervised OCL-PDS with active learning and optimized pseudo-labeling.
  - In the Amazon-WPDS dataset, our approach outperformed the ER-FIFO method